

Prodigal Social - Multi-Agent Social Media Company

Live Demo Record

Fully local execution via Ollama. No hosted LLM APIs.

Repo: https://github.com/yashieeeeeee/social_ai_company

Model: qwen2.5:3b (Q4_K_M, 1.9 GB) - one model, all 8 agents.

Hardware: Intel i5-1240P, 8 GB RAM, CPU-only, Windows 64-bit.

Date: 2026-09-14

This document walks through the system exactly as it ran on the candidate machine: setup from a clean clone, the test suite, the model proof, a full two-week campaign in deterministic fallback mode, and the same campaign driven live by the local model. Every number below is a real measured output, and every figure is a screenshot of the actual run.

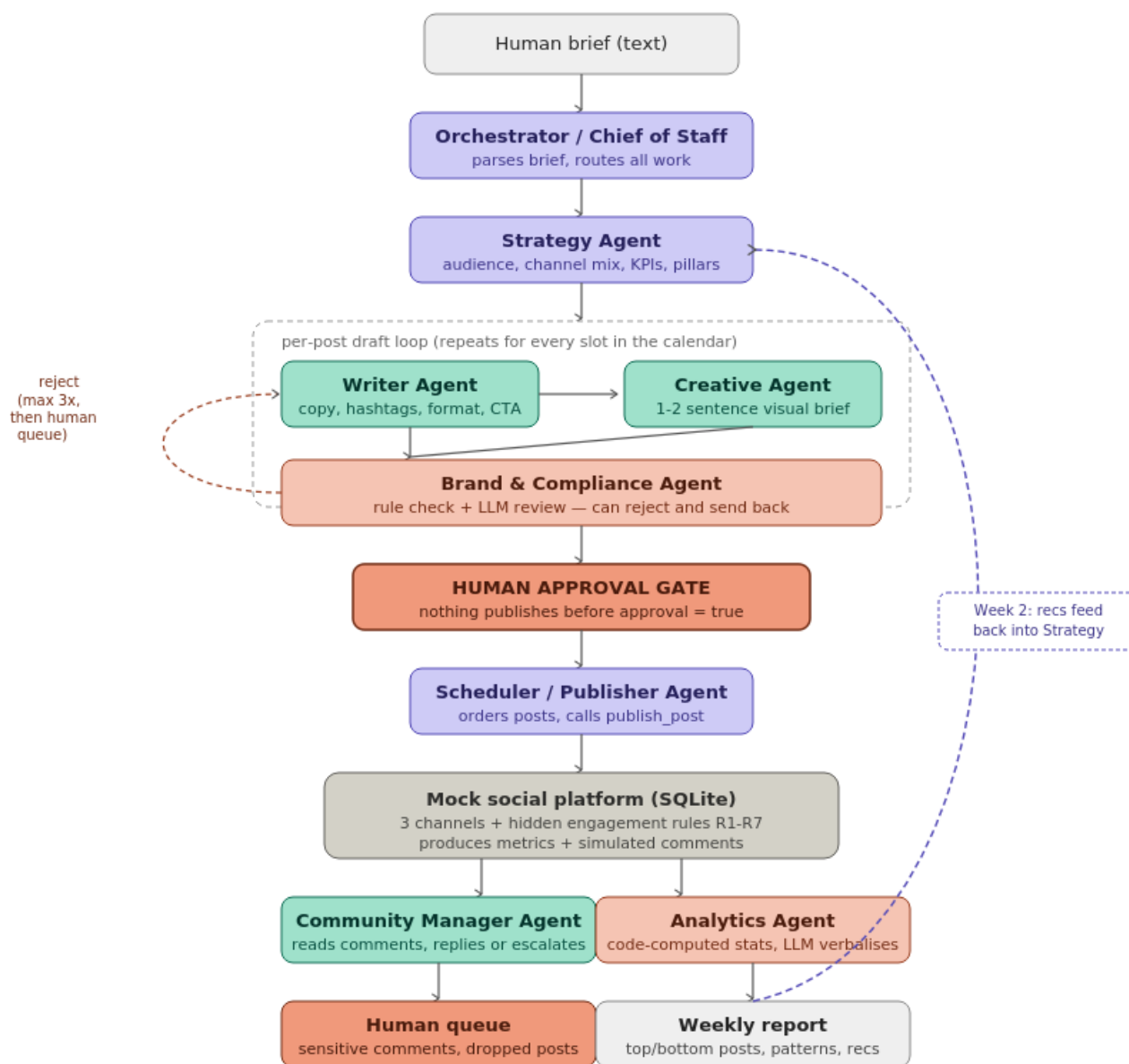
1. What the system does

A human brief goes in. The Orchestrator parses it, Strategy plans a 12-post week across three mock channels (buzz, forum, pro), Writer and Creative draft each post, Compliance reviews with a 3-strike cap into a human queue, a CLI approval gate must say yes before anything publishes, the Scheduler publishes to a SQLite mock platform with hidden engagement rules R1-R7, the Community Manager answers comments or escalates, and Analytics produces a numeric weekly report whose recommendations feed Week 2. Full agent trace lands in logs/trace.jsonl.

2. Architecture and message flow

Agents never call each other. Every handoff is `bus.send()` into a shared SQLite messages table mirrored to `trace.jsonl`, routed by CampaignRunner.

Prodigal Social — agent architecture and message flow



All handoffs above go through `bus.send(from, to, type, payload)` — a shared SQLite messages table mirrored to `logs/trace.jsonl`. Agents never call each other directly.

3. Setup from a clean clone (Figure 1)

Cloned into an empty folder, installed one dependency (requests), ran with --no-llm. No model weights, services, or setup scripts needed; SQLite initialises itself on first run.

```
PS C:\Users\Acer> git clone https://github.com/yashieeeee/social_ai_company.git fresh-test-clone
Cloning into 'fresh-test-clone'...
remote: Enumerating objects: 65, done.
remote: Counting objects: 100% (65/65), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 65 (delta 23), reused 57 (delta 19), pack-reused 0 (from 0)
Receiving objects: 100% (65/65), 47.43 KiB | 1.39 MiB/s, done.
Resolving deltas: 100% (23/23), done.
PS C:\Users\Acer> cd fresh-test-clone
PS C:\Users\Acer\fresh-test-clone> pip install -r requirements.txt
Requirement already satisfied: requests in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from -r requirements.txt (line 1)) (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (3.5.1)
Requirement already satisfied: idna<4,>=2.5 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (3.19)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (2.7.0)
Requirement already satisfied: certifi<=2023.5.7 in c:\users\acer\appdata\local\programs\python\python314\lib\site-packages (from requests->r requirements.txt (line 1)) (2026.7.22)
[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Acer\fresh-test-clone> python run.py --brief-file demo_brief.txt --auto-approve --no-llm
```

Figure: git clone + pip install in a fresh folder. Proves zero-context setup.

4. Test suite (Figure 2)

27 tests, mocked LLM (no model needed): hidden-rule spot checks, JSON repair and fallback behaviour, compliance retry cap, gate-fails-closed, streak detector, per-post error containment.

```
.
-----
Ran 27 tests in 1.084s
OK
PS C:\Users\Acer\fresh-test-clone> |
```

Figure: Ran 27 tests in ~1s - OK on the fresh clone.

5. Local model proof (Figure 3)

qwen2.5:3b pulled and listed locally (1.9 GB, sha256-verified). The only network call in the codebase is POST localhost:11434/api/generate.

```
OK
PS C:\Users\Acer\fresh-test-clone> ollama pull qwen2.5:3b
pulling manifest
pulling 5ee4f07cdb9b: 100% 1.9 GB
pulling 66b9ea09bd5b: 100% 68 B
pulling eb4402837c78: 100% 1.5 KB
pulling b5c0e5cf74cf: 100% 7.4 KB
pulling 161dde4c9cd: 100% 487 B
verifying sha256 digest
writing manifest
success
PS C:\Users\Acer\fresh-test-clone> ollama list
NAME ID SIZE MODIFIED
qwen2.5:3b 357c53fb659c 1.9 GB 2 minutes ago
PS C:\Users\Acer\fresh-test-clone> |
```

Figure: ollama pull + ollama list on the demo machine.

6. Demo run A - reproducible baseline, fallback mode (Figures 4-6)

Same brief both weeks: budget espresso machine for students, playful, no over-promising. Week 1: 32,584 impressions, 1,427 comments, 3 escalations. Analytics found evening +106% reach (n=5/7) and x5.4 question-comment lift, so Week 2 shifted to evening slots: 42,812 impressions (delta +31%). Usage ledger 128 calls / 128 fallbacks: every output from deterministic templates, which is what makes it bit-for-bit reproducible for any evaluator without a GPU.

```
===== CAMPAIGN cmp-cli-1 (12 posts, round 1) =====
Objective: awareness | Audience: price-sensitive 18-25 students
[buzz d1 19:00] (howto/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[buzz d2 09:00] (story/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[buzz d3 19:30] (poll/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee,students
[buzz d5 03:00] (meme/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee,students
[forum d1 12:30] (howto/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee,students
[forum d2 19:00] (story/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee,students
[forum d4 09:00] (poll/question) Student espresso for under a rupee a cup? price on a budget - what matters ... | coffee,students
[forum d6 18:30] (meme/none) Student espresso for under a rupee a cup. dorm-life on a budget, built for ... | coffee,students
[pro d1 09:00] (howto/question) Student espresso for under a rupee a cup? taste on a budget - what matters ... | coffee
[pro d3 19:00] (story/none) Student espresso for under a rupee a cup. price on a budget, built for dorm... | coffee
[pro d4 12:30] (poll/question) Student espresso for under a rupee a cup? dorm-life on a budget - what matt... | coffee
[pro d6 08:30] (meme/none) Student espresso for under a rupee a cup. taste on a budget, built for dorm... | coffee
[gate] --auto-approve: approved
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON call failed attempt 1 (orchestrator): --no-llm mode
2026-09-14 16:34:48,456 prodigal.llm WARNING: LLM JSON fallback (orchestrator): ollama-call-failed: --no-llm mode

[publish] 12 posts live.
```

Figure: Week-1 campaign: 12 posts, gate approved, 12 live.

```
[week1] imp=32584 comments=1427 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+106% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x5.4 comments)
```

Figure: Week-1 report lines with the two numeric recs.

```
--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *           approve

LLM usage: {'calls': 128, 'tokens_in': 0, 'tokens_out': 0, 'fallbacks': 128}
```

Figure: Week 2 delta +31%, bus handoffs, usage ledger.

7. Demo run B - live local model, the headline (Figures 7-8)

Same pipeline, model doing the writing: 134 calls, 26,310 tokens in, 8,885 out, 0 fallbacks. The trace caught a live compliance reject-then-rewrite-then-approve on camera. Week 1: 21,167 impressions, 542 comments. Week 2: 16,583 (delta -22%). The regression is understood, not hidden: model drafts collapsed to one format (11/12 howto) with zero question-endings, so Week 2 kept the evening gain but ate the R5 novelty decay penalty while harvesting none of R2. A second interactive run (human typed y at the gate) gave 149 calls, 2 fallbacks (analytics timeouts), W1 20,322 vs W2 16,043 (delta -21%). Honest numbers beat flattering ones; the loop works, the model has a diversity failure with a named mechanism.

```
Windows PowerShell
[publish] 12 posts live.
[week1] imp=21167 comments=542 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+131% imp)
REC: end 1 in 2 forum/buzz posts with a genuine question (x1.9 comments)
[week2] imp=16583 (delta -22%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *          approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer
orchestrator --[compliance_attempt]--> *          reject
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *          approve

LLM usage: {'calls': 134, 'tokens_in': 26310, 'tokens_out': 8885, 'fallbacks': 0}
PS D:\dev\prodigal social> |
```

Figure: Live-model campaign: 11 posts, week1/week2 lines, 134 calls 0 fallbacks.

```
Windows PowerShell
rs_BehindTheScenes
[forum d4 09:00] (poll/Share your favorite espresso story in the comments!) Did you know espresso was first brewed in Italy in the 16th century? It's n... | espresso,coffeefun,italiancul
ture
[forum d6 18:30] (meme/Give it a shot and let me know how it turns out!) Hey coffee lovers! Ever wished your morning cup was as perfect as that fanc... | coffeebrewing,morningcoffee,coff
eaddicts
[pro d1 09:00] (howto/Watch now to learn more about your favorite brew!) Ever wondered how your perfect cup of espresso is made? Today at 9 AM, join... | espresso,coffeemaking,barista
[pro d3 19:00] (howto/What's your favorite way to enjoy espresso?) Did you know? Espresso is named after espresso coffee, which is a quick way... | FunFacts,EspressoLovers,CaffeineCult
ure
[pro d4 12:30] (howto/Try it out and let me know how it goes!) Got a coffee craving? Here's how to make the perfect cup at home. Start wit... | coffee,brewing,homebrew
[pro d6 08:30] (howto/Learn More) Ever wondered how your favorite café keeps its espresso shots so perfect? L... | espresso,coffeemachine,barista
[y]approve [s]re-plan strategy [w]rewrite posts [N]quit > y

[publish] 12 posts live.
2026-09-14 18:08:10,743 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:08:10,831 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week1] imp=20322 comments=516 escalated=3
REC: shift buzz/forum posts from 09:00/03:00 to 18:00-21:00 (+123% imp)
2026-09-14 18:25:42,818 prodigal.llm WARNING: LLM JSON call failed attempt 1 (analytics): HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
2026-09-14 18:25:43,159 prodigal.llm WARNING: LLM JSON fallback (analytics): ollama-call-failed: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=120)
[week2] imp=16043 (delta -21%)

--- TRACE (first 25) ---
orchestrator --[llm_call]--> orchestrator
orchestrator --[brief_parsed]--> strategy
strategy --[llm_call]--> orchestrator
strategy --[plan_ready]--> orchestrator
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[reject]--> writer
orchestrator --[compliance_attempt]--> *          reject
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *          approve
writer --[llm_call]--> orchestrator
writer --[draft_ready]--> creative
creative --[llm_call]--> orchestrator
creative --[brief_ready]--> compliance
compliance --[llm_call]--> orchestrator
compliance --[approve]--> scheduler
orchestrator --[compliance_attempt]--> *          approve

LLM usage: {'calls': 149, 'tokens_in': 28007, 'tokens_out': 8340, 'fallbacks': 2}
PS C:\Users\Acer\fresh-test-clone> |
```

Figure: Interactive gate (typed y), reject rows in trace, 149 calls 2 fallbacks.

8. Reproduce everything

```
git clone https://github.com/yashieeeeeee/social_ai_company.git
pip install -r requirements.txt
python -m unittest discover -s tests
python run.py --brief-file demo_brief.txt --auto-approve --no-llm
python run.py --brief-file demo_brief.txt --auto-approve (needs Ollama)
```

9. Honest scorecard vs planted rules R1-R7

Found: R1 prime-time (acted on, measured lift), R2 question boost (needs draft variance). Noted: R6 CTA effect. Missed by design: R3/R4/R7 are censored by Compliance before publish (rejected drafts logged as counterfactual post_dropped messages); R5 needs a 3-day same-format streak the rotating calendar never forms (detector built and unit-tested on synthetic streaks). With the live model, Week 2 regressed via format collapse into R5 - documented above with the mechanism, not excused.